

Perio Voice AI - Technical Overview

Speech-to-Text

Uses **Deepgram Nova-3** for real-time streaming speech-to-text.

Clinical Understanding

Uses the **Oxlo API** for clinical language understanding and command interpretation.

Audio Pipeline

Browser microphone audio is captured, converted to **16 kHz mono PCM**, and streamed continuously to a **FastAPI** backend over **WebSockets**.

Latency Strategy

Continuous streaming instead of uploading complete recordings minimizes latency and enables near real-time chart updates.

Transcript Processing

A normalization layer corrects common speech recognition errors in spoken numbers and periodontal terminology before sending the transcript to the Oxlo API.

Sequencing Approach

A state machine tracks the current tooth, surface, and probing site, automatically advances after valid depth triplets, and supports voice commands such as bleeding, missing tooth, implant, undo, and tooth override.

Engineering Challenges Solved

- Fixed frontend/backend synchronization issues.
- Improved payload hydration and state management.
- Used immutable React state updates for reliable UI rendering.
- Corrected SVG rendering so measurements appear immediately on the correct tooth.
- Optimized WebSocket communication and audio streaming for better responsiveness and stability.

Current Limitations

- Occasional speech recognition errors during rapid dictation or in noisy environments.
- Clinical accuracy can be further improve using benchmark datasets and additional tuning of transcript normalization and command parsing.